

Large Language Models (LLMs) based on the Transformer architecture have revolutionized Natural Language Processing (NLP). The self-attention mechanism allows these models to weigh the contextual importance of words in a sentence, regardless of their positional distance. This paper proposes a novel optimization technique for the attention matrix, reducing computational overhead by 40% while maintaining 95% accuracy on standard benchmark datasets like GLUE and SuperGLUE. Our methodology replaces standard dense layers with sparse tensor representations during the pre-training phase.